

Project Proposal

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September 19, 2026

Abstract

High-resolution spectroscopic surveys provide precise stellar parameters and chemical abundances, but their samples contain only a small fraction of the stars with astrometry and low-resolution spectrophotometry from *Gaia*. Building directly on the *Gaia* XP chemical-tagging framework developed by Kane et al., we retrain and adapt a multi-output neural network to transfer labels from APOGEE DR17 to *Gaia* DR3 BP/RP XP spectra for a targeted study of the Sagittarius system. The model predicts T_{eff} , $\log g$, $[\text{Fe}/\text{H}]$, $[\text{N}/\text{O}]$, and $[\text{Al}/\text{Fe}]$, together with a source-dependent variance for each label. We use a heteroscedastic Gaussian likelihood to model observational scatter and Monte Carlo dropout to estimate model uncertainty. This would not only add to the existing catalogs but also provide a way to test machine learning algorithms for use in astronomy and astrophysics.

1 Introduction

The chemical and dynamical properties of stars preserve information about the formation of the Milky Way and its accreted satellites. In particular, metallicities and elemental-abundance ratios can distinguish stellar populations, constrain enrichment histories, and identify debris associated with disrupted dwarf galaxies. The Sagittarius dwarf spheroidal galaxy (hereafter Sgr) is a nearby satellite undergoing tidal disruption by the Milky Way and therefore provides a valuable laboratory for studying chemical evolution during an ongoing accretion event.

Detailed abundance measurements are commonly obtained from medium- and high-resolution spectroscopic surveys. The Apache Point Observatory Galactic Evolution Experiment (APOGEE) provides precise near-infrared stellar labels for hundreds of thousands of stars, including a spectroscopic subset of the Sgr system [abdurrouf2022](#). However, the number of probable Sgr members identified using *Gaia* astrometry is much larger than the number with existing spectroscopic metallicities and abundances. The recent Sgr core and M54 membership catalogue of [toguchitani2026](#), for example, contains approximately 1.4×10^5 Sgr core sources and approximately 2×10^3 M54 sources, while spectroscopic labels are available for only a subset.

Gaia Data Release 3 (DR3) provides internally calibrated low-resolution BP/RP spectrophotometry, commonly referred to as XP spectra, for roughly 220 million sources [gaiadr3,deangeli2023](#). These data are represented by basis-function coefficients and contain broad spectral information across the optical and near-infrared wavelength range. Several recent studies have demonstrated that data-driven models can infer stellar parameters, metallicities, and selected abundance information from XP data [e.g., [li2024](#), [khalatyan2024](#)]. The information available at XP resolution is nevertheless entangled with temperature, surface gravity, extinction, survey selection, and correlations among chemical labels. Consequently, both careful validation and a quantitative assessment of predictive uncertainty are required before applying a trained model to a distinct population such as Sgr.

Of particular relevance, [kane2025](#) developed a heteroscedastic multilayer perceptron that used *Gaia* BP/RP coefficients to infer T_{eff} , $\log g$, $[\text{Fe}/\text{H}]$, $[\text{N}/\text{O}]$, and $[\text{Al}/\text{Fe}]$ on the APOGEE abundance scale. Their network used 108 normalized BP/RP coefficients, two hidden layers of 128 neurons, dropout with $p = 0.2$, and 100 stochastic forward passes. They used the resulting high- $[\text{N}/\text{O}]$ and high- $[\text{Al}/\text{Fe}]$ predictions to identify 878 new candidate globular-cluster-origin stars in the Galactic field, reporting an overall classification accuracy of approximately 99% and a false-positive rate of approximately 7%. This work demonstrated both the chemical-tagging potential of XP spectra and the importance of abundance-specific validation at their low spectral resolution.

For every target label, the network will predict both a mean and an input-dependent variance. We additionally use dropout at inference time to estimate the sensitivity of the prediction to the learned network parameters gal2016,kendall2017,kane2025. We will then apply the model to the Sgr core and M54 membership samples of toguchitani2026.

The primary goals of this study are to:

1. determine the precision with which T_{eff} , $\log g$, $[\text{Fe}/\text{H}]$, $[\text{N}/\text{O}]$, and $[\text{Al}/\text{Fe}]$ can be transferred from APOGEE to stars with *Gaia* XP spectra;
2. test whether the predicted uncertainties are statistically calibrated;
3. identify the magnitude, color, stellar-parameter, and XP-coefficient domains in which the model provides reliable interpolation rather than extrapolation; and
4. use the expanded label sample to characterize the metallicity and abundance structure of the Sgr core and M54.

The paper is organized as follows. Section 2 describes the APOGEE–*Gaia* training sample and the Sgr membership catalogue. Section ?? presents the network architecture, loss function, training procedure, and uncertainty model. Section ?? validates the predictions on held-out stars. Section ?? presents the application to Sgr and M54. We discuss the astrophysical interpretation and limitations in Section ??, and summarize our conclusions in Section ??.

2 Data

2.1 Gaia DR3 XP Spectra

Gaia DR3 publishes BP and RP mean spectra as coefficients multiplying internally calibrated basis functions deangeli2023. For each source, we concatenate the retained BP and RP coefficients into a feature vector

$$\mathbf{x}_i = (c_{i,1}^{\text{BP}}, \dots, c_{i,N_{\text{BP}}}^{\text{BP}}, c_{i,1}^{\text{RP}}, \dots, c_{i,N_{\text{RP}}}^{\text{RP}}), \quad (1)$$

where i indexes the star. The input construction is adapted from kane2025. In their preprocessing, each of the 55 BP and 55 RP coefficients is divided by the first coefficient of the corresponding arm; the two resulting unit-valued first coefficients are then removed, leaving 108 predictors. Our final pipeline uses $N_{\text{XP}=108}$ input features.

We retrieve XP coefficients using the *Gaia* archive table `gaia_source` crossmatched to the APOGEE labels. Sources with non-finite coefficients are removed. We additionally investigate the dependence of model performance on G magnitude, $G_{\text{BP}} - G_{\text{RP}}$ color, extinction, and XP goodness-of-fit statistics.

2.2 APOGEE DR17 Labels

The reference labels are drawn from APOGEE DR17 abdurrouf2022. Following the label-transfer problem defined by kane2025, we use the following five targets:

$$\mathbf{y}_i = (T_{\text{eff}}, \log g, [\text{Fe}/\text{H}], [\text{N}/\text{O}], [\text{Al}/\text{Fe}])_i. \quad (2)$$

The exact catalogue columns are `TEFF`, `LOGG`, `FE_H`, `NO`, and `AL_FE`. The label denoted $[\text{N}/\text{O}]$ throughout this paper must be verified against the data-reading function before analysis and publication.

2.3 Proposed Methodology

We propose to develop and train a multilayer perceptron (MLP) neural network to predict missing stellar labels for stars in the *Gaia* catalogue. The network will be trained as a supervised regression model using stars for which both *Gaia* observations and independently measured stellar labels are available. The trained model will then be applied to stars lacking these labels in order to infer their stellar parameters and chemical abundances.

The input features will consist of the available *Gaia* measurements, while the target variables will be the corresponding known stellar labels in the training sample. Prior to training, the data will be

cleaned, normalized, and divided into training, validation, and test sets. The validation set will be used to monitor model performance during training, while the test set will provide an independent assessment of the network's predictive accuracy.

The MLP will consist of multiple fully connected hidden layers with nonlinear activation functions. Model parameters will be optimized through backpropagation by minimizing an appropriate regression loss function. We will investigate the effects of network architecture, learning rate, batch size, and other hyperparameters on model performance.

The model will be implemented primarily using `PyTorch`, with `scikit-learn` used for data pre-processing, dataset splitting, and performance evaluation. Additional machine-learning tools may be incorporated as necessary. Additionally, implementation of the `huggingface` gpu accelerator will allow us to shorten learning times and test various network parameters for an optimal pipeline.

References